Lakmali Nadeesha Kumari

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Education

University Of Kentucky,USA — Expected graduation date: April 2027
Ph.D. (Postgraduate) in Electrical and Computer Engineering | Link to all courses — *GPA: 3.8/4.00*
Relevant Courses: Image Processing, Stochastic Systems, Algorithm Design, Bayesian Machine Learning, Advanced Computational Methods for Biomedical Imaging, Process Monitoring and Machine Learning, Mathematical Introduction to Deep Learning, Deep Learning Based Model Predictive Control

University Of Peradeniya, Sri Lanka — Washington Accord degree program
B.Sc. in Engineering in Electrical and Electronic Engineering | Link to all courses

Mahamaya Girls' College, Kandy, Sri Lanka
General Certificate of Education (GCE) Advanced Level (2012) — *Ranked in the top 1% among ∼250,000 candidates*

Experience

Multimedia Information Analysis Laboratory (MIA Lab) — Aug. 2022 - Present
Ph.D. Student/ Researcher — *AI in Medicine, Machine Learning, Computer Vision, Active Learning, Data Mining, Python*

- Developed difficulty-aware segmentation pipelines for class-imbalanced gigapixel histopathology images using foundation models, without extra estimators or training stages.
- Identified Alzheimer's progression subtypes from regional amyloid and NFT biomarkers, revealing subtype-specific genetic associations driving spatial spreading patterns.
- Built a self-supervised active learning framework with adaptive Gaussian mixture models for skin lesion segmentation, reducing annotation cost.
- Created a feature-extracted gaze behavior dataset using computer vision techniques to support early detection of Autism Spectrum Disorder.

Amazon – Merch on Demand, Seattle, WA, USA — Jun. 2024 – Aug. 2024, May 2025 – Aug. 2025

Applied Scientist (L5) Intern-Summer 2025 — *Generative AI, Computer Vision, Deep Learning*

- Designed and implemented an end-to-end generative AI pipeline for automated, context-aware product visualization leveraging diffusion models, LLM-based quality evaluation, and LoRA-based refinement on multi-GPU cloud infrastructure.

Applied Scientist (L5) Intern-Summer 2024 — *Multi-Modal Classification, Deep Learning, Computer Vision*

- Built a multi-modal, multi-label classification system with an automated active learning framework and LLM-based labeling to categorize large-scale product data, significantly outperforming keyword-based baselines.

Augmented Intelligence for Smart Manufacturing (AISM) Lab — Aug. 2021 - Aug. 2022
Ph.D. Student/ Researcher — *Machine Learning, Machine Prognosis, Predictive Maintenance, Python, Matlab*

- Developed a Conditional Invertible Neural Network framework for Lithium-Ion Battery remaining useful life prediction with precise degradation modeling.

Synopsys Sri Lanka-Software company — April. 2018 - Aug. 2021
Applications Engineer II (Service Letter), (Awards) — *Linux, Verilog, SystemVerilog, VHDL, Bash, Perl, HTML*

- Component owner for SpyGlass Analyzer; validated static verification tools for identifying critical RTL design issues.
- Validated industry-leading clock domain crossing (CDC) and reset domain crossing (RDC) verification solutions.
- Validated design constraint tools to eliminate design flaws and prevent costly re-spins.

Teaching

Graduate Teaching Assistant, University of Kentucky, USA — Jan. 2022 - Dec. 2023

Instructor (Teaching Assistant), University of Peradeniya, Sri Lanka — Nov. 2017 - April. 2018

Undergraduate Internships

Electrical and Electronic Engineering undergraduate trainee at Ceylon Electricity Board — Oct. 2016 - Jan. 2017

Electrical and Electronic Engineering undergraduate trainee at Sri Lanka Telecom (PLC) — Oct. 2015 - Jan. 2016

Publications and Projects

International Publications

- **R.M.L.N. Kumari**, G.A.C.T. Bandara, Sen-ching Cheung, **"A Warmer Start to Active Learning with Adaptive Gaussian Mixture Models for Skin Lesion Segmentation"**,IEEE International Conference on Image Processing (ICIP) 2025,DOI: 10.1109/ICIP55913.2025.11084666. **Link to the paper**
- **R.M.L.N. Kumari**, P. Wang, **"Efficient stochastic parametric estimation for lithium-ion battery performance degradation tracking and prognosis"**, Journal of Manufacturing Systems, 2024, Elsevier. DOI: 10.1016/j.jmsy.2024.03.017. **Link to the Journal**
- **R.M.L.N. Kumari**, G.A.C.T. Bandara, and Dr. Maheshi B. Dissanayake, **"Sylvester Matrix Based Similarity Estimation Method for Automation of Defect Detection in Textile Fabrics"**, Journal of Sensors, vol. 2021, 2021. DOI: 10.1155/2021/6625421. **Link to the Journal**
- G.A.C.T. Bandara, **R.M.L.N. Kumari**, Dr. D.H.S. Maithripala and Dr. D.A.A.C. Ratnaweera, **"Vehicle-Fixed-Frame Adaptive Controller and Intrinsic Nonlinear PID Controller for Attitude Stabilization of a Complex-Shaped Underwater Vehicle"**, Journal of Mechatronics and Robotics, vol. 4, 2020. DOI: 10.3844/jmrsp.2020.254.264. **Link to the Journal**

Preprints / Manuscripts in Preparation

- **R.M.L.N. Kumari**, Sen-ching Cheung, **"Learning Class Difficulty in Imbalanced Histopathology Segmentation via Dynamic Focal Attention"**, [Under Review]–Medical Image Computing and Computer-Assisted Intervention (MICCAI) 2026.
- G.A.C.T. Bandara, **R.M.L.N. Kumari**, Sen-ching Cheung, **"Not All Steps Matter: Temporally Adaptive Spectral Control for Concept Removal in Diffusion Models"**, Target venue: Conference on Neural Information Processing Systems (NeurIPS), 2026.

Research Symposium

- **R.M.L.N. Kumari**, and Dr. Sen-ching Cheung, **"SemiSAM_Med: Class-Imbalanced Active Learning in Medical Image Segmentation via a Semi-supervised SAM Model with Domain-Adapted Transfer Learning"**, The Electrical and Computer Engineering Department Research Symposium -2024, Lexington, USA. **Link to the poster**
- **R.M.L.N. Kumari**, and Dr. Sen-ching Cheung, **"Eyes Speak Louder: Computer Vision and Machine Learning Insights into Autism Spectrum Behaviors"**, The Commonwealth Computational Summit 2023: Artificial Intelligence, University of Kentucky's Center for Computational Sciences, USA. **Link to the poster**
- **R.M.L.N. Kumari** and Dr. P. Wang, **"Modeling and Prediction of Lithium-Ion Battery Degradation with CINN for Parameter Estimation"**, The Engineering Dean's Advisory Council Meeting-2023, Lexington, USA. **Link to the poster**
- **R.M.L.N. Kumari**, and Dr. Sen-ching Cheung, **"SAL-SMIS: Self-Supervised Active Learning for Semi-Supervised Medical Image Segmentation"**, The Electrical and Computer Engineering Department Research Symposium -2023, Lexington, USA. **Link to the poster**
- **R.M.L.N. Kumari**, G.C.D Bandara, W.A.S.S. Bandara, Dr. J. V. Wijayakulasooriy, and Dr. H. A. C. Dharmagunawardhana, **"Rotational Invariant Image Registration for PCB Fault Detection"**, The Electrical and Electronic Engineering Research and Project Symposium (EEERaPS)-2017, Peradeniya, Sri Lanka. **Link to the poster**
- **R.M.L.N. Kumari**, G.C.D Bandara, W.A.S.S. Bandara, Prof. M. Fernando, and Dr. S. Kumara, **"Analyses the inrush current of power transformers"**, The Electrical and Electronic Engineering Research and Project Symposium (EEERaPS)-2017, Peradeniya, Sri Lanka.(Poster session presented)

Extra-Curricular Activities

Assistant Director, the graduate Society of Women Engineers (GradSWE), University of Kentucky	2023 - 2024
Vice-President (VP) membership in Synopsys Lanaka Toastmaster Club	2020
Secretary of Synopsys Sport Club	2019
Senior member of basketball team of University of Peradeniya. *University colours*	2014 - 2017
Senior member of hockey team of University of Peradeniya.*University colours*	2014 - 2017
Member of Sri Lanka National University Women's basketball team	2017
Member of Kandy District (Youth Clubs) Women's basketball team	2017
Volunteer - 2017 12th IEEE International Conference on Industrial and Information Systems (ICIIS)	2017
Active member of the Electrical and Electronic Engineering Society (EEES), University of Peradeniya	2015 - 2017
National level volunteer mathematics and science teacher for GCE Ordinary Level student in 'Arunella' Social Project	2014
Participated in the training program on "Developing Leadership Qualities and Positive Thinking"	2013
Associate Member of Institution of Engineers, Sri Lanka	2016 - Present

Skills

Programming:	Python, MATLAB, R, C/C++, Bash, SQL, Verilog, SystemVerilog, VHDL, HTML
ML / DL:	PyTorch, TensorFlow, Hugging Face, CLIP, Diffusion Models, LoRA/PEFT
Tools & Infra:	AWS (EC2, Bedrock, SageMaker), Docker, VS Code, Git, Linux, Multi-GPU Training, LaTeX